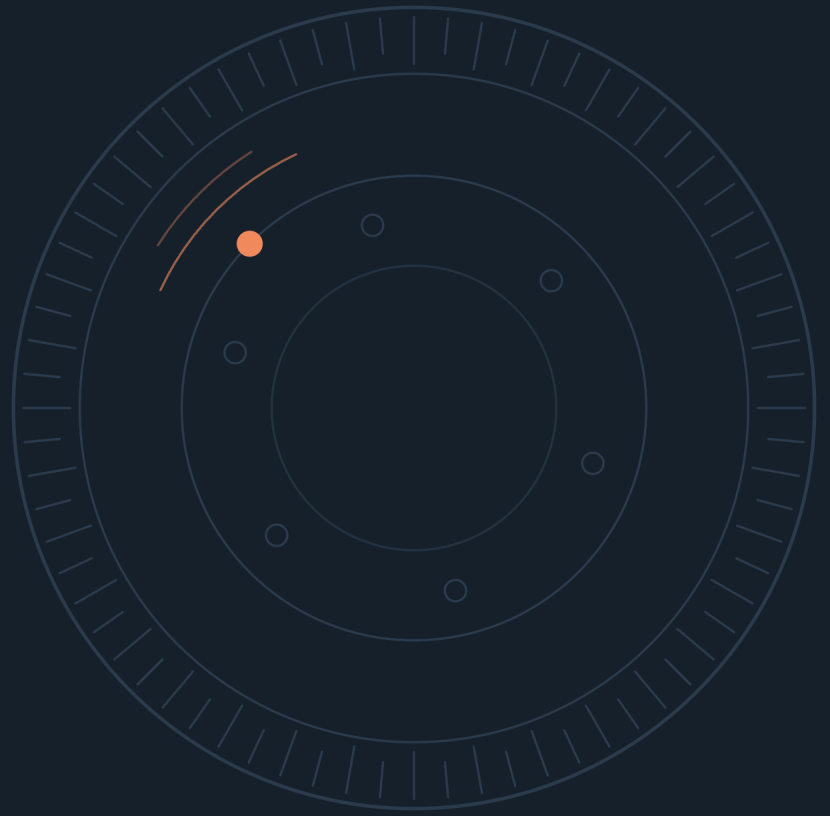
Westline Fleet Tyre Performance

An Evidence-Led Assessment of Tyre Pressure, Monitoring and Operational Practices



PREPARED BY

Ritika Raj Parekh

CHALLENGE

Humaniti AI Build Challenge

DATA PERIOD

1 June – 15 July 2026

The client’s four questions, answered

Could the Karwar failure have been foreseen?

Not a day ahead. On the day, yes: it split from its twin from 03:15 and passed a 10 PSI gap at 03:40, five hours before it fell below 70 PSI.

Can one alarm name one tyre?

A twin comparison is a strong candidate: inner against outer on the same hub. It stays quiet through heat and cooling and flagged only two real problems in six weeks.

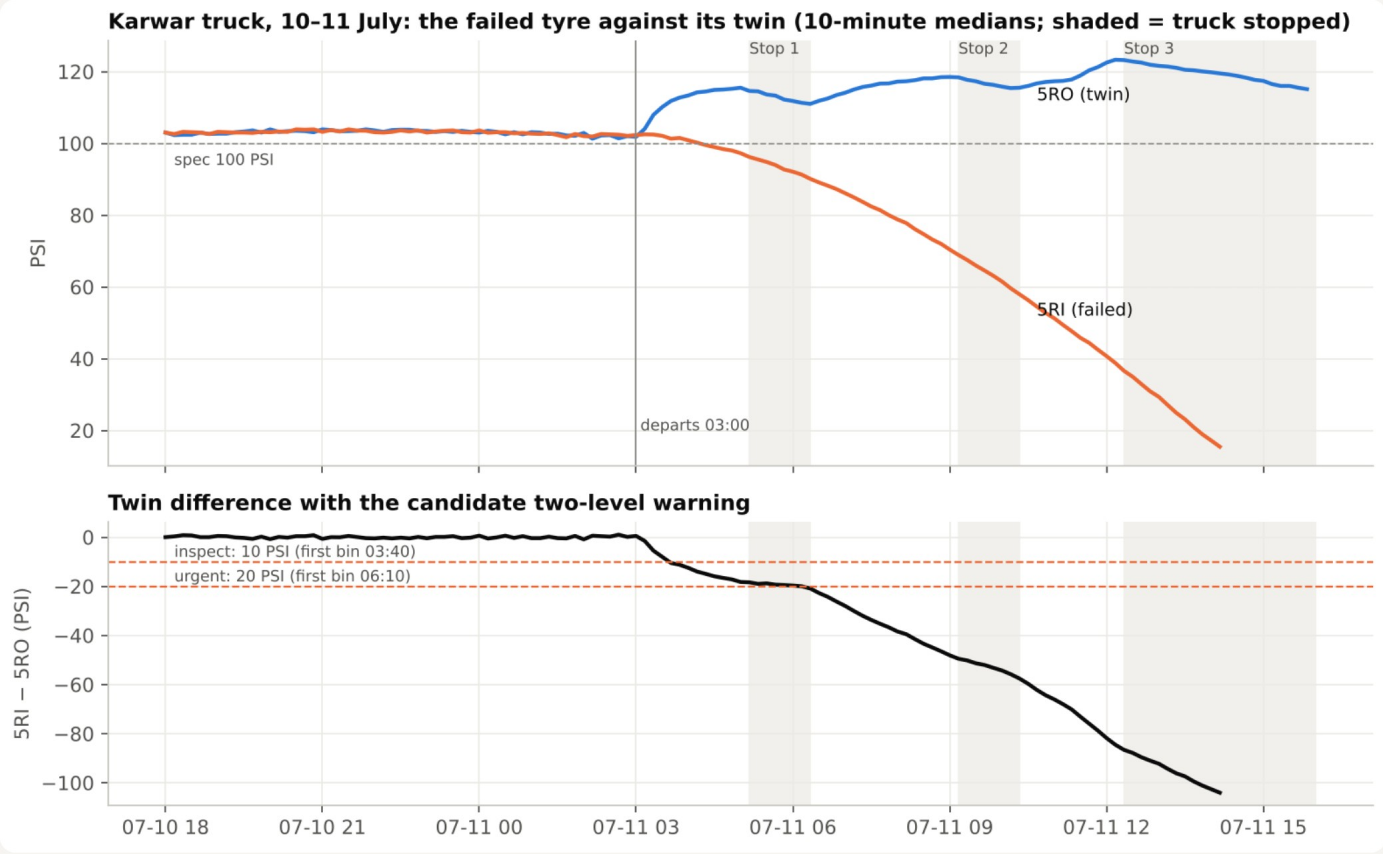
Are the Hubballi roads eating the tyres?

Not mainly. Hubballi fills hot tyres in the afternoon to the cold card value, so its trucks start every morning 10 to 12 PSI under spec.

Why “nothing but alarms” on the Goa truck?

All fourteen tyres cool together in the afternoon and recover. That is temperature, not leaks, and it happens on every truck.

Karwar 5RI gave hours of warning on the day, none the day before



10 July night: both tyres 102 to 103 PSI, parked. No earlier day showed even a 5 PSI gap.

03:40 gap passes 10 PSI, before Stop 1.

06:10 gap passes 20 PSI; tyre still about 92 PSI.

09:03 below 70 PSI, before the Hubballi stop.

14:19 last reading, 14.6 PSI, while parked near Kittur.

Findings F9, F12. The physical failure time is unknown (“by two o’clock”, second-hand), so I give no single lead time.

A candidate one-tyre warning: compare each tyre with its twin

10 PSI

Inspect

Inner and outer on one hub differ by 10 PSI for 20 minutes. Six weeks: 21 episodes from just two conditions.

20 PSI

Urgent

One episode in six weeks: the Karwar failure, from 06:10.

8.1

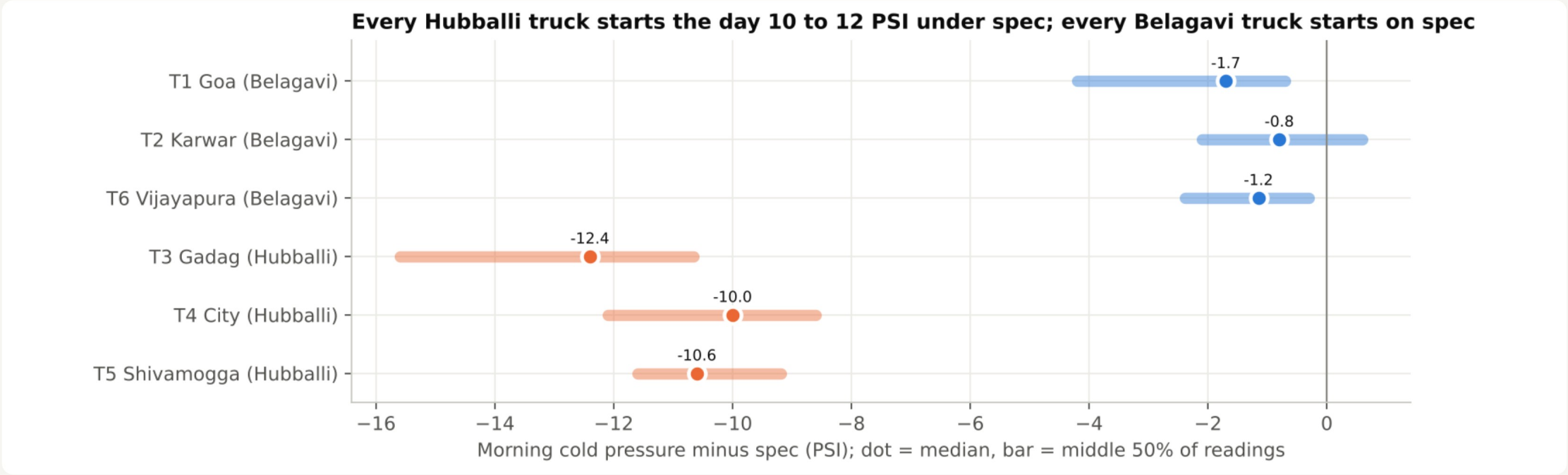
Largest gap in Goa cooling

Twins cool together, so the afternoon events never reach 10 PSI.

Why twins: they carry the same load at the same speed on the same road in the same weather, so a difference isolates one tyre. It needs a reset band (the 20 Gadag episodes are one imbalance re-crossing the line), and it is tested on one failure only. Axles 1 to 3 have no twin and need another comparison.

Findings F13, F17. All 24 twin pairs, 10-minute medians on paired readings, 1 June to 15 July.

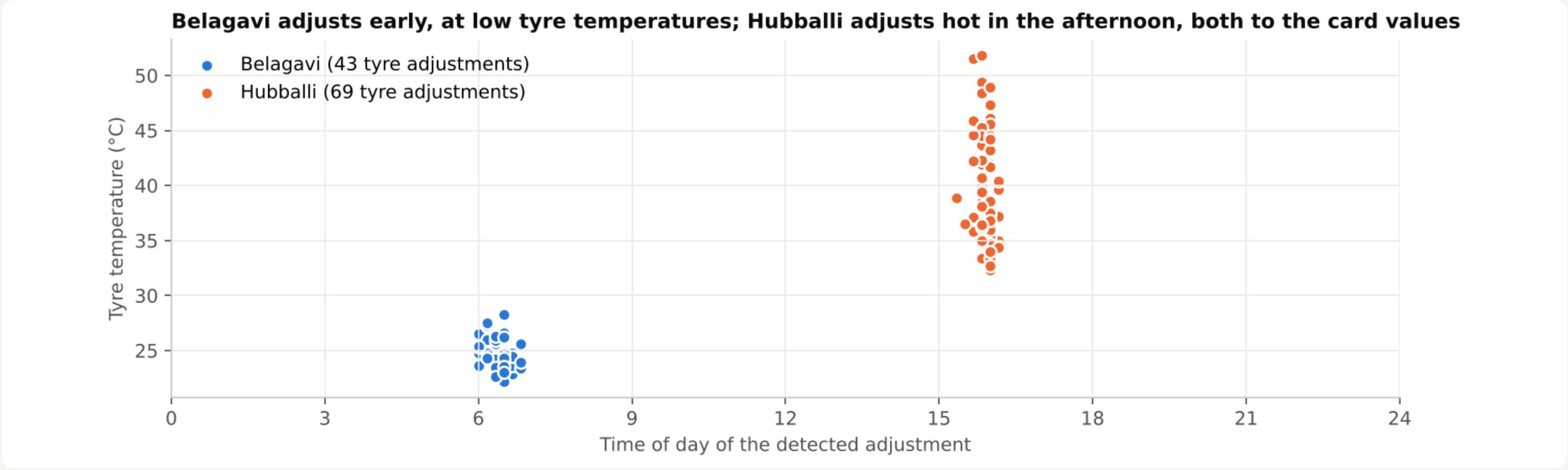
Every Hubballi truck starts the day 10 to 12 PSI under spec



Cold readings after at least 3 hours parked, 3,062 tyre readings. At least 10 PSI under: 51 to 82% of Hubballi readings, 0 to 0.6% of Belagavi's. The gap is on every axle, so it is something the depot does, not individual tyres.

Finding F14. Ramesh: "ninety, ninety-one on the drive tyres." He is right.

The cause is when they fill, not the roads

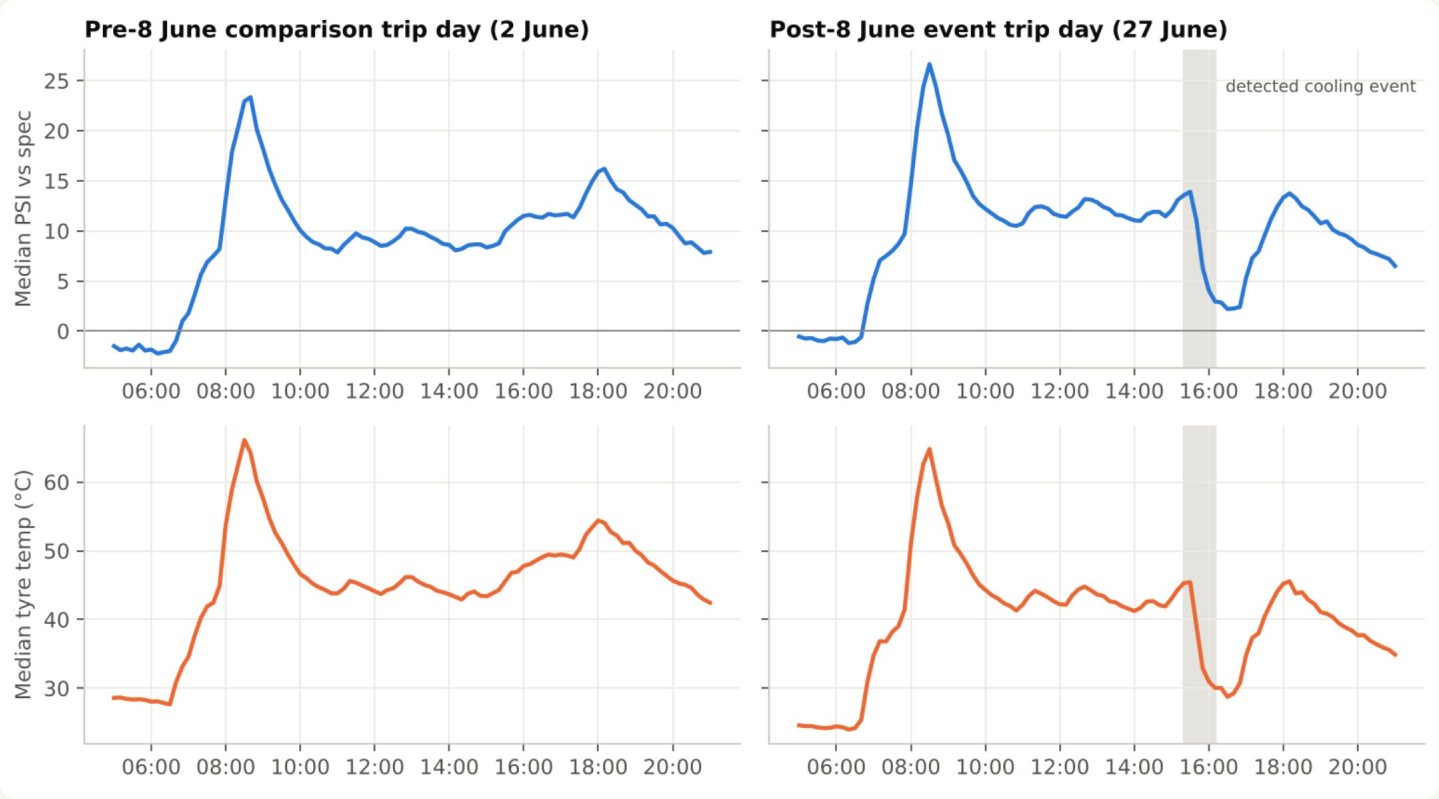


Hot fills. Hubballi sets hot tyres to the cold card value. A tyre set to 100 PSI at 40 °C reads about 90 once cold. All 69 Hubballi adjustments were 6 to 18 PSI under the next morning; all 43 Belagavi adjustments within 2.5 PSI.

Roads, a small part at most. Between fills Hubballi tyres lose 1 to 2 PSI per 30 days more than Belagavi's: months to make a gap that appears overnight.

Findings F15, F16. Adjustments detected as a 3 PSI or larger step while parked, with temperature steady.

The Goa afternoon drops are the whole truck cooling



All 14 tyres drop together in every one of 13 events.

0.66 to 0.73 PSI per °C as temperature falls 12 to 18 °C: the rate cooling predicts.

Recovers to within 0.4 to 1.6 PSI as tyres warm (9 of 13 measurable).

From 9 June, 13:00 to 17:00, on all six trucks.

Finding F17. Fits Ramesh's afternoon rain; no weather data to prove it. Alarms themselves need the TyreSense alert log.

What to do now

Action	Why	Finding
Fill tyres cold at Hubballi	Hot afternoon fills leave every truck 10 to 12 PSI under by morning	F14, F15
Trial the twin-difference warning	10 PSI inspect, 20 PSI urgent, with a reset band; caught Karwar on the day	F12, F13
Inspect four tyres	Goa 4LI, Karwar 3LO, Gadag 5RO, City 5RI lose air several times faster than the spec's 3 PSI a month	F16
Replace the Shivamogga 2RO sensor	Battery at 1%, missed readings doubling	F18
Fix the data feed	Duplicated 12-hour overlap, wrong registration on one file, C/D side codes mislabelled	F1–F3
Get the alert log	Needed to see which events actually alarmed and who received them	F17

Five questions for Ramesh, in order

#	Question	What the answer lets us do
1	Who receives TyreSense alerts, at what thresholds, and can we have the June–July alert log?	Tell which alarms fired, and route the twin warning to someone who can call the driver
2	At Hubballi, who fills the tyres, at what time, with which gauge?	Confirm the hot-fill cause and move fills to the cold morning check
3	Karwar 5RI: what was fitted at the roadside, when was 0901 fitted, and what failed?	Correct the register; judge whether a twin warning catches the next one
4	Are Goa 4LI, Karwar 3LO, Gadag 5RO and City 5RI known to leak?	Inspect them this week if not
5	Can TyreSense supply fields the export lacks: speed, altitude, sensor IDs, ignition?	Explain battery jumps, tell engine-on from parked, test Anmod descent heat

What the data cannot tell us yet

One failure. A warning rule cannot be validated, or its false-alarm rate estimated, on a single case.

No weather, load or ignition data. Rain, weight and engine state are inferred or unknown.

Records disagree. Register and transcript differ on the Karwar replacement, and the data dictionary has the C/D side codes the wrong way round.

No alert log. I can show what the tyres did, not which alarms fired or who saw them.

Six weeks. Slow leaks near the spec's 3 PSI a month are at the edge of what this period can resolve.

Partial inspections. Only the Goa truck has two tread readings; two trucks have none.

Full evidence, confidence and open questions for each of 18 findings: Westline_Findings_Full.html